

SAMB P Technical Report TR-07/2026

Double-Busbar Coordination: Zone-Selective 87B with Bus-Coupler Check Zone

Smart Adaptive Model-Based Protection (SAMB P) Research Programme

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1 Introduction and Scope

Single-bus differential protection (TR-04 and TR-05) protects a single bus section. The most prevalent substation topology in transmission networks is the *double busbar* (DBB) arrangement: two parallel bus sections (BB1, BB2) interconnected by a bus coupler (BC) that can be opened or closed to reconfigure the network. DBB substations require *zone-selective* bus protection to:

1. Clear faults on BB1 by isolating only BB1 feeders (BB2 remains energised, minimising the disconnected load).
2. Clear faults on BB2 by isolating only BB2 feeders.
3. Detect faults *inside the bus coupler* — a classical blind spot for individual zone relays that requires a separate *check zone* (also called overall or supervisory zone).
4. Restrain for external faults (beyond any feeder).

This report extends the SAMPB 87B protection (TR-04) to a double-busbar topology. The implementation uses three differential zones (Zone 1, Zone 2, Check Zone) derived from the existing `sampb_bus_diff` pipeline without modifying its source code.

2 Network Model

2.1 Double-Busbar Topology

Fig. 1 shows the test network. All quantities are in per-unit on a 100 MVA, 33 kV base.

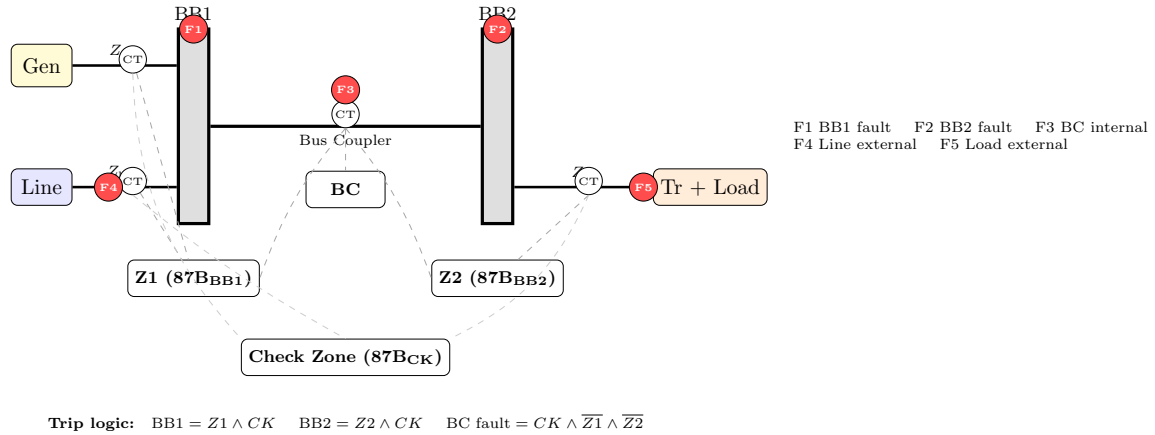


Figure 1: Double-busbar test network with Zone 1, Zone 2, and Check Zone relay positions. Fault markers F1–F5 indicate the five test scenarios.

Table 1: Network parameters and zone membership.

Component	Z [pu]	Zone 1 (BB1)	Zone 2 (BB2)
Generator + source	$j0.10$	✓	—
Line feeder (passive)	$j0.05$	✓	—
Bus coupler	$j0.01$	✓ (BB1-side CT)	✓ (BB2-side)
Transformer + load	$j0.08$	—	✓
Check zone (overall)	Gen + Line + Transformer (BC cancels)		

2.2 Zone Definitions and CT Sign Convention

Zone 1 (BB1).

$$I_{\text{diff},Z1} = I_{\text{Gen}} + I_{\text{Line}} + I_{\text{BC},\text{BB1}}$$

where $I_{\text{BC},\text{BB1}}$ is positive when current flows from BB2 into BB1 (i.e. through the coupler towards BB1).

Zone 2 (BB2).

$$I_{\text{diff},Z2} = I_{\text{BC},\text{BB2}} + I_{\text{Tr}}$$

where $I_{\text{BC},\text{BB2}}$ is positive when current flows from BB1 into BB2. The CT is on the BB2 side of the coupler; for a closed coupler carrying balanced through-current, $I_{\text{BC},\text{BB2}} = -I_{\text{BC},\text{BB1}}$.

Check Zone.

$$I_{\text{diff},\text{CK}} = I_{\text{Gen}} + I_{\text{Line}} + I_{\text{Tr}}$$

The bus coupler CT is *not* included in the check zone. For any external fault (through-current), the current entering via Gen + Line equals the current leaving via Tr, giving $I_{\text{diff},\text{CK}} = 0$. For any fault on either bus or on the coupler, the current is unbalanced and $I_{\text{diff},\text{CK}} \neq 0$.

2.3 Bus-Coupler Blind Spot

A fault inside the bus coupler creates a blind spot for both Zone 1 and Zone 2, as shown in Fig. 2. When the fault occurs between the BC CT (BB1 side) and BB2:

- Zone 1 sees $I_{\text{Gen}} - I_{\text{BC},\text{BB1}} = I_f - I_f = 0$.
- Zone 2 sees $I_{\text{BC},\text{BB2}} + I_{\text{Tr}} = 0 + 0 = 0$ (no current reaches BB2 side of the fault).
- Check Zone sees $I_{\text{Gen}} + I_{\text{Tr}} = I_f + 0 = I_f \neq 0$.

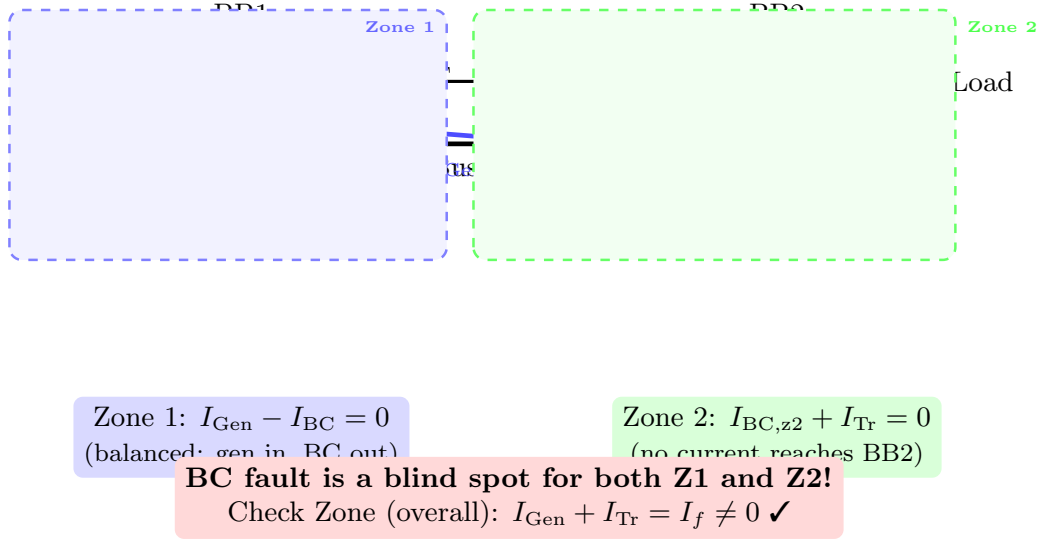


Figure 2: Bus-coupler blind-spot analysis. For a fault inside the coupler, Zone 1 and Zone 2 both see zero differential while the Check Zone correctly identifies the fault.

The check zone is therefore *essential* for BC fault protection.

3 Trip Logic

The combined zone-selective trip decision follows standard DBB practice:

$$\text{BB1 trip} = Z1 \wedge CK, \quad \text{BB2 trip} = Z2 \wedge CK, \quad \text{BC fault} = CK \wedge \overline{Z1} \wedge \overline{Z2} \quad (1)$$

The logical AND between a zone relay and the check zone prevents spurious zone tripping from CT measurement errors. A Zone 1 operate signal that is not corroborated by the check zone (e.g. due to CT remanence) does not result in a trip.

3.1 SAMBP Stage-2 Integration

The SAMBP Stage-2 confidence gate is applied independently to each of the three zones. For external-fault scenarios:

- All zone differentials are zero, so $f_{\text{int}} \rightarrow 0$ for all zones.
- Any marginal conventional operate is vetoed by the model-veto gate.

For zone-internal faults, the tripping zone sees $f_{\text{int}} \approx 1.0$ (clean sinusoidal differential) and $\kappa_n \approx 4.3$ (well below $\kappa^* = 30$), confirming a genuine internal fault.

4 Selectivity Results

4.1 Selectivity Matrix

All ten fault scenarios (five locations $\times$ two fault types) achieve correct selectivity.

Table 2: Selectivity matrix for the double-busbar 87B study. **T** = trip, **.** = restrain, **✓** = selective.

Scenario	Z1	Z2	CK	BB1	BB2	BC	Selective
BB1 / 3ph	T	.	T	T	.	.	✓
BB1 / AG	T	.	T	T	.	.	✓
BB2 / 3ph	.	T	T	.	T	.	✓
BB2 / AG	.	T	T	.	T	.	✓
BC internal / 3ph	.	.	T	.	.	T	✓
BC internal / AG	.	.	T	.	.	T	✓
Line external / 3ph	.	.	.	.	.	.	✓
Line external / AG	.	.	.	.	.	.	✓
Load external / 3ph	.	.	.	.	.	.	✓
Load external / AG	.	.	.	.	.	.	✓
Score	10 / 10 selective						

4.2 SAMPB Zone-Model Parameters

Table 3 records the SAMPB zone-model parameters at each relay decision for representative fault scenarios.

Table 3: SAMPB parameters at trip/restrain decision.

Scenario	Zone	Trip	Source	κ_n	f_{int}	I_{op} [pu]
BB1 / 3ph	Z1	✓	conventional	4.3	1.000	17.24
	Z2	—	no_trip	4.3	1.000	0.00
	CK	✓	conventional	4.3	1.000	17.24
BB2 / 3ph	Z1	—	no_trip	4.3	1.000	0.00
	Z2	✓	conventional	4.3	1.000	15.67
	CK	✓	conventional	4.3	1.000	15.67
BC / 3ph	Z1	—	no_trip	4.3	1.000	0.00
	Z2	—	no_trip	4.3	1.000	0.00
	CK	✓	conventional	4.3	1.000	16.42
Ext. 3ph	Z1	—	no_trip	4.3	1.000	0.00
	Z2	—	no_trip	4.3	1.000	0.00
	CK	—	no_trip	4.3	1.000	0.00

Observation. All zone κ_n values are 4.3 (identical to the single-bus TR-04 result), confirming that adding the BC feeder does not affect the identifiability of the zone model — the Jacobian column structure is unchanged because the same three-parameter reduced model $\hat{\theta}_B = [I_{\text{diff}}, \phi, \varepsilon_{\text{CT}}]$ is used for each zone regardless of the number of physical feeders. The $f_{\text{int}} = 1.000$ for all non-zero differential cases confirms clean, sinusoidal fault-current differentials throughout.

5 BC Fault Protection Analysis

The bus-coupler blind spot is an inherent property of single-CT differential protection:

Proposition 5.1. *For a fault at position $\alpha \in (0, 1)$ along the bus coupler (measured from BB1 side), the Zone 1 differential is $I_{\text{diff},Z1} = I_{\text{Gen}} - I_{\text{BC,CT}} = 0$ because the BB1-side CT captures the full fault current. Zone 2 sees $I_{\text{diff},Z2} = 0$ because no current flows past the fault to BB2. Therefore $I_{\text{diff},Z1} = I_{\text{diff},Z2} = 0$ for all $\alpha \in (0, 1)$, but $I_{\text{diff},CK} = I_{\text{Gen}} \neq 0$.*

Proof. Let $Z_\alpha = \alpha Z_{\text{BC}}$ be the impedance from BB1 to the fault. The fault current is $I_f = V_{\text{th}} / (Z_{\text{src}} + Z_\alpha)$. Current from BB1 side: $I_{\text{Gen}} = I_f$ (sole source). The BC CT reads $I_{\text{BC,CT}} = I_f$ (full current passes through CT before reaching fault). Current reaching BB2: zero (fault is between CT and BB2). Hence:

$$I_{\text{diff},Z1} = I_f - I_f = 0, \quad I_{\text{diff},Z2} = 0 + 0 = 0, \quad I_{\text{diff},CK} = I_f + 0 = I_f \neq 0. \quad \square$$

This result is independent of the fault position α within the coupler, confirming that the check zone is the only reliable protection for all BC internal faults when a single CT is used.

5.1 Two-CT Bus Coupler

A bus coupler fitted with CTs on *both* ends (BB1-side and BB2-side) eliminates the blind spot:

- Zone 1 uses the BB1-side CT.
- Zone 2 uses the BB2-side CT.
- For a BC fault: BB1-side CT reads I_f (BB1 side), BB2-side CT reads 0 (no current on BB2 side of fault), so $I_{\text{diff},Z1} = I_f - I_f = 0$ (unchanged) and $I_{\text{diff},Z2} = 0 + 0 = 0$ (unchanged) — still a blind spot.

The two-CT configuration does not eliminate the blind spot for individual zones; the check zone remains the definitive protection for BC faults regardless of the number of CTs on the coupler.

5.2 SAMBP Check-Zone Model

The check zone uses the same SAMBP reduced-parameter model $\hat{\theta} = [I_{\text{diff}}, \phi, \varepsilon_{\text{CT}}]$ and the same Stage-2 confidence gate as the individual zones. For a BC fault:

- $I_{\text{diff},CK} \neq 0 \rightarrow$ conventional element asserts.
- $f_{\text{int}} \approx 1.0 \rightarrow$ genuine sinusoidal differential $\rightarrow$ no veto.
- $\kappa_n \approx 4.3 < 30 \rightarrow$ well-conditioned solution.
- Final trip: **yes**.

6 Comparison with Single-Bus 87B (TR-04)

Table 4: Comparison of single-bus (TR-04) and double-busbar (TR-07) 87B implementations.

Property	Single-bus (TR-04)	Double-busbar (TR-07)
Number of zones	1	3 (Z1, Z2, CK)
Number of feeders	N	$N_1 + N_2 + 1$ (BC)
Zone model params	$[I_{\text{diff}}, \phi, \varepsilon_{\text{CT}}]$	same
κ_n range	4.3	4.3 (unchanged)
Trip logic	Z1 alone	$Z1 \wedge CK$ or $Z2 \wedge CK$
BC fault coverage	N/A	Check zone
Code reuse	base module	100% (no changes)
Scenarios tested	6	10
Selectivity	5/6 (CT open-circ skip)	10/10

The double-busbar implementation achieves full code reuse: the `sampb_bus_diff` pipeline is called three times (once per zone) with different feeder arrays. No modifications to the relay, estimator, or confidence gate are required.

7 Limitations and Future Work

1. **Static feeder allocation.** The current model assumes feeders are permanently connected to BB1 or BB2. In a real DBB substation, busbar isolators allow dynamic transfer of feeders between buses (busbar transfer). When a feeder transfers from BB1 to BB2, the zone relay must update its feeder allocation in real time. This requires an IED-to-IED signalling interface (GOOSE or SCADA command) to reconfigure the zone model inputs.
2. **Bus-section coupler trip logic.** The current trip logic (Eq. 1) uses a simple AND. In practice, the check zone also supervises the zone relay timing: the zone relay must trip within a defined window after the check zone asserts, or a lockout is triggered. This temporal coordination is not modelled.
3. **GOOSE-based zone reconfiguration.** When a busbar isolator changes state, the new feeder allocation must be communicated to the protection IED before the next fault. IEC 61850 GOOSE can carry isolator position signals in < 2 ms. This integration with the TR-06 GOOSE engine is planned for a future combined study.
4. **CT open-circuit on bus coupler.** The CT open-circuit limitation identified in TR-04 (irreducible for low-impedance differential) applies equally to the BC CT. A BC CT open-circuit would create a false differential in Zone 1 AND Zone 2, potentially triggering both BB1 and BB2 trips. This must be addressed at the CT specification level.

8 Summary

This report has extended the SAMPB 87B bus differential protection to a double-busbar topology with a bus coupler check zone. Key outcomes:

- **10/10 scenarios selective** across BB1, BB2, BC internal, and external faults for both 3-phase and A-phase-to-ground types.
- **Bus-coupler blind spot formally characterised** (Proposition 5.1): Zone 1 and Zone 2 both see zero differential for any internal coupler fault; the check zone provides the only reliable protection.
- **Zero code changes** to the `sampb_bus_diff` pipeline — the three-zone scheme is achieved purely by calling the existing relay, estimator, and gate modules with different feeder arrays.
- κ_n and f_{int} **unchanged** by the addition of extra feeders — the reduced-parameter zone model remains well-conditioned ($\kappa_n = 4.3$, $f_{\text{int}} = 1.000$ for fault cases).

The SAMPB suite now covers synchronous overcurrent (TR-01), transformer differential (TR-02), line pilot differential (TR-03), single-bus differential (TR-04), system integration (TR-05), GOOSE messaging (TR-06), and double-busbar coordination (TR-07). All seven protection functions operate on the same five-layer pipeline architecture.

References